Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/26/2024
PI Name:	Ruben, Josh		
Project Title:	Virtual Environment Oculometrics in Low-Earth-Orbit (VEO in LEO)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	josh@z3vr.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	919-279-2752
Organization Name:	Z3VR		
PI Address 1:	1301 Fannin St		
PI Address 2:	#2440		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77003	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2021	**End Date:**	01/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	SpaceX		
Flight Assignment:	SpaceX Inspiration4 NOTE: End date changed to 01/31/2024 per E. Urquieta/TRISH (Ed., 10/5/23) NOTE: End date changed to 04/30/2023 per E. Urquieta/TRISH (Ed., 8/14/23) NOTE: End date changed to 01/31/2024 per E. Urquieta/TRISH (Ed., 6/28/23) NOTE: End date updated to 04/30/2024 (from end date of 05/31/2024) per E. Urquieta/TRISH (Ed., 6/5/23) NOTE: End date changed to 01/31/2024 per E. Urquieta/TRISH (Ed., 1/30/23) NOTE: End date changed to 04/30/2023 per E. Urquieta/TRISH (Ed., 6/2/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stone, Leland Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX16AO69A-EXP0003		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EXPAND Project Upcoming exploration class mission crews will have to endure long periods in microgravity, and then land while reloading in +Gz—making readiness-to-perform assessments vital for crew safety and mission success. There is, therefore, an urgent need to develop and validate rapid, non-invasive visual and vestibular testing systems for use in spaceflight operational settings. On Earth, our 5-minute oculometric test can detect and distinguish mild neural impairment caused by head trauma, sleep loss, and low-dose alcohol (Liston et al., 2017; Stone et al., 2019) and can even predict performance in manual control and highly skilled baseball tasks (Chen et al., 2021). Under our ongoing "Oculometric Cognition Testing and Analysis in Virtual Environments" (OCTAVE) grant, we are translating our laboratory system into an operationally useful, standalone, lightweight, low-power, self-proctored Virtual Reality (VR) system. With this new grant, we will accelerate development and incorporate vestibular testing into our test battery for use on the Inspiration4 flight. We will validate a 15-minute testing session focused on the immediate and short time-constant neural adaptation of visual and vestibular responses known to affect crew performance. Previous studies have shown that directional tuning of visual processing is immediately and significantly altered by changing gravity on Earth (e.g., in supine testing, +Gx centrifugation, and spaceflight). Due to fundamental physics, the otoliths immediately stop signaling head tilts in microgravity, leading to errors in gaze stabilization, ocular alignment, motion sickness, and sensorimotor discoordination. This drives a neural adaptation process with an increased role for canal signals that takes a few days to recalibrate vestibular ocular and spinal reflexes to enable effective sensorimotor performance in microgravity. The mirror process occurs upon landing, with immediate perceptual illusions, motion sickness, and postural instability, followed by a re-adaptation that can take several days, with considerable inter-subject variability. Spaceflight also causes otolith signals to be reinterpreted as linear accelerations instead of tilt. Upon reloading, we expect to see maladaptive smooth horizontal reflexive eye movements in response to roll head tilts. Given that re-adaptation takes a few days, with high inter-subject variability, it is important to have a system to assess re-adaptation progress before performing high-risk tasks (e.g., the first post-landing extravehicular activity (EVA)). Validation (even if only tentative) of such a system during the Inspiration4 flight would greatly benefit future exploration efforts. Beyond performance assessment and sensorimotor monitoring, VR will be invaluable to NASA as a tool for training, recreation, and behavioral health support. In addition, it will expand commercial spaceflight capabilities in passenger screening, research, and customer satisfaction. References Liston DB, Wong LR, Stone LS. Oculometric assessment of sensorimotor impairment associated with TBI. Optom Vis Sci. 2017 Jan;94(1):51-59. Stone LS, Tyson TL, Cravalho PF, Feick NH, Flynn-Evans EE. Distinct pattern of oculomotor impairment associated with acute sleep loss and circadian misalignment. J Physiol. 2019 Sep;597(17):4643-4660. Chen R, Stone LS, Li L. Visuomotor predictors of batting performance in baseball players. J Vis. 2021 Mar 1;21(3):3.Chen et al., 2021.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The research project aimed to enhance health monitoring using innovative technologies, with specific benefits tailored for both space and Earth environments: Space: 1) Continuous Health Monitoring: The project intended to enable unobtrusive, continuous health monitoring of astronauts during long-duration missions by validating eye tracking technology in microgravity. 2) Development of Countermeasures: It aimed to inform the development of targeted countermeasures to mitigate the adverse effects of microgravity on visual function and cognitive performance. 3) Support for Future Missions: The research was designed to support future exploration missions by advancing our understanding of maintaining crew health and performance beyond low Earth orbit. Earth: 1) Non-Invasive Monitoring: The project succeeded in enabling continuous, non-invasive monitoring of cognitive performance and well-being through the integration of advanced eye tracking technologies into consumer wearables. 2) Personalized Wellness Recommendations: It developed personalized wellness recommendations based on individual oculometric data to optimize performance and manage stress. 3) Medical Applications: The research facilitated medical applications of eye tracking for early detection of neurological disorders, treatment monitoring, and diagnosis of conditions affecting visual function and cognitive performance. We were not able to fly the device in microgravity, so we were not able to achieve the proposed impacts listed for space. However we have developed a system that achieves all relevant Earthbound research impacts.
Bibliography Type:	Description: (Last Updated:)